

OSSP (Operating Systems And Systems Programming) - 25CS2104E

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SEC-8

SKILL-5:

Aim: To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

PROCEDURE :

1. Single quote test :

```
siddinapujith07@DESKTOP-FDHCAIT:~$ mkdir skill5
siddinapujith07@DESKTOP-FDHCAIT:~$ cd skill5
siddinapujith07@DESKTOP-FDHCAIT:~/skill5$ nano! quotes.c
siddinapujith07@DESKTOP-FDHCAIT:~/skill5$ gcc quotes.c -o quotes
siddinapujith07@DESKTOP-FDHCAIT:~/skill5$ ./quotes
Enter a quoted command: '$HOME hello world'

Input: '$HOME hello world'
Type: Single Quoted String
Literal content preserved.
Variable expansion ignored.
Parsing completed successfully.
```

2. Double quote with \$HOME :

```
siddinapujith07@DESKTOP-FDHCAIT:~/skill5$ ./quotes
Enter a quoted command: "$HOME hello world"

Input: "$HOME hello world"
Type: Double Quoted String
Spaces preserved.
Variable found: $HOME
Expanded value: /home/siddinapujith07
Parsing completed successfully.
```

3. Double quote without variable :

```
siddinapujith07@DESKTOP-FDHCAIT:~/skill5$ ./quotes
Enter a quoted command: 'hello world'

Input: 'hello world'
Type: Double Quoted String
Spaces preserved.
No variable found.
Parsing completed successfully.
```

Code:

```
GNU nano 8.7.1          quotes.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main()
{
    char input[200];
    printf("Enter a quoted command: ");
    fgets(input, sizeof(input), stdin);
    input[strcspn(input, "\n")] = '\0';
    printf("\nInput: %s\n", input);
    if (input[0] == '\'' )
    {
        printf("Type: Single Quoted String\n");
        printf("Literal content preserved.\n");
        printf("Variable expansion ignored.\n");
    }
    else if (input[0] == '"' )
    {
        printf("Type: Double Quoted String\n");
        printf("Spaces preserved.\n");
        char *pos = strstr(input, "$HOME");
        if (pos != NULL)
        {
            printf("Variable found: $HOME\n");
            printf("Expanded value: %s\n", getenv("HOME"));
        }
        else
        {
            printf("No variable found.\n");
        }
    }
    else
    {
        printf("No quotation detected.\n");
    }
}
```

```
}  
  
printf("Parsing completed successfully.\n");  
  
return 0;  
}
```

Observation :

1. Single quotes preserved the enclosed content literally and prevented variable expansion.
2. Double quotes preserved spaces while allowing \$HOME variable expansion.
3. Quoted strings were correctly identified and parsed.
4. Different quoted inputs and edge cases were tested successfully.

Result :

The program successfully implemented **single-quote and double-quote handling**, preserving literal content and spaces as required while supporting variable expansion inside double quotes. The quoted inputs were parsed and validated successfully.